

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE End Semester Examination (Regular) – December 2019 Branch: B.Tech in Mechanical Engineering Subject Name with Code: Heat Transfer [BTMEC 501] Max Marks: 60 Sem: V Date: 09/12/2019 Duration: 3 hr.		
Instructions to the Students: 1. Solve ANY FIVE questions out of the following six questions. 2. Each question carries 12 marks. 3. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of each question. 4. Use of non-programmable scientific calculator is allowed. 5. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)
Q. 1	Solve the following:	Marks
A)	Define overall heat transfer coefficient. How is it related to thermal resistance? Write down equation for total thermal resistance for composite hollow cylinder made up of two layers (with thermal conductivity k_1, k_2 with thickness b_1, b_2) and convective boundary conditions on either sides of composite cylinder with convective heat transfer coefficients h_1, h_2 . Assume some notation for radii like r_1, r_2, r_3 .	(CO-2) 6
B)	A composite wall of 1m^2 surface area is constructed of two layers. The first layer is of 1 cm thick steel ($k = 45 \text{ W/m K}$) and the second layer of 10 cm thick fiberglass insulation ($k = 0.035 \text{ W/m K}$). Determine: a) thermal resistance of the composite wall b) overall heat transfer coefficient.	(CO-1) 6
Q.2	Solve the following:	
A)	Derive equation for critical radius of insulation for a cylinder. A 2 mm diameter electric wire at 46°C is covered by 0.5 mm thick plastic insulation ($k = 0.03 \text{ W/m K}$). The insulation of the wire is exposed a medium at 10°C with convective heat transfer coefficient of $20.0 \text{ W/m}^2 \text{ K}$. Determine the critical insulation thickness. Will plastic insulation dissipate max heat?	(CO-2) 6
B)	Write equations for Biot number & Fourier number. Aluminium sphere weighing 5.5 kg and initially at a temperature of 290°C is suddenly immersed in a fluid at 15°C . The convective heat transfer coefficient is $58 \text{ W/m}^2\text{K}$. Estimate the time required to cool the aluminium to 95°C . Use lumped capacity method for calculation. Assume following property values for aluminium: $\rho = 2700 \text{ kg/m}^3$ $k = 205 \text{ W/m K}$ $C_p = 900 \text{ J/kg K}$	(CO-2) 6
Q. 3	Solve the following:	
A)	Sketch laminar and turbulent boundary layers (BL) for flow over a flat plate. Also, show velocity profiles within the BL in the two regions: a) in the laminar region b) in the	(CO-4) 6

	turbulent region. Assume uniform velocity profile on the upstream side of the plate. State the value of Re at transition.		
B)	Air at 60°C and atmospheric pressure flows over a thin flat plate (<i>one side</i>) which is 1 m wide and 2 m in length. The free stream air velocity is 1 m/s. Calculate a) thickness of velocity boundary layer at a distance $x = 1.5$ m and b) total drag force on the plate. Use following values of fluid properties: $\nu = 18.96 \times 10^{-6} \text{ m}^2/\text{s}$, $\rho = 1.06 \text{ kg/m}^3$ Use following equations for thickness of velocity boundary layer and drag coefficient, respectively: $\frac{\delta}{x} = \frac{5}{\sqrt{Re_x}} \quad \text{and} \quad C_f = 1.328 Re_L^{-0.5}$	(CO-4)	6
Q.4	Solve the following:		
A)	Water at 50°C enters 1.5 cm diameter tube of a heat exchanger. Assume velocity of water at mean temperature as 1 m/s. The tube surface is maintained at 90°C. Calculate the exit water temperature if the length of tube is 2 m. Assume following properties of water at mean temperature (<i>neglect variation in properties with temperature</i>): $\begin{aligned} \mu &= 489.2 \times 10^{-6} \text{ kg/(m.s)} \\ \rho &= 984.4 \text{ kg/m}^3 \\ k &= 0.656 \text{ W/m K} \\ c_p &= 4178 \text{ J/kg K} \end{aligned}$ Use following correlations: $f = 0.079 Re^{-0.25}$ $Nu_D = \frac{(f/2)[Re_D - 1000].Pr}{1 + 12.7 \left[\frac{f}{2} \right]^{0.5} (Pr^{2/3} - 1)}$	(CO-4)	6
B)	A 15 cm diameter horizontal iron pipe with 1 m length is exposed to saturated steam at 100°C on inside and still air at 20°C on outside. Calculate the convective heat transfer rate from the outer surface of pipe and compare it with radiant heat transfer rate if the surrounding surfaces (imaginary) are at 20°C. Assume outer surface of the pipe as black. Use following correlation for horizontal pipe: $Nu = 0.48 Ra^{0.25}$ Use following values of properties of air at 60°C: $\rho = 1.06 \text{ kg/m}^3 \quad \mu = 20.1 \times 10^{-6} \text{ N.s/m}^2, \quad k = 0.029 \text{ W/m K} \quad C_p = 1005 \text{ J/kg K}$	(CO-4)	6
Q. 5	Solve the following:		
A)	In a double-pipe counter-flow heat exchanger 10,000 kg/h of an oil having a specific heat 2095 J/kg.K is cooled from 80°C to 50°C by 8000 kg/h of cooling water at 25°C. Assume overall heat transfer coefficient of 300 W/m²K and specific heat of water as 4180 J/kg.K. a) Determine LMTD b) Determine the heat exchanger area.	(CO-5)	6
B)	Draw the pool boiling curve & identify different boiling regimes. Define critical flux & show corresponding point on pool boiling curve.		6

Q. 6	Solve the following:		
A)	Consider a 20 cm diameter spherical ball at 800 K suspended in air. Assuming the ball closely approximates a blackbody, determine: a) The total blackbody emissive power b) The amount of radiant energy emitted by the ball in 5 min. c) The monochromatic blackbody emissive power at a wavelength of 3 micrometers.	(CO-6)	6
B)	State various shape factor relations (algebra) in radiation heat transfer.	(CO-6)	6
	*** End ***		